Data acquisition (APIs, Scraping, public datasets)

Rennan Marujo

Application Programming Interface (API)

How does a restaurant works?

Client

Waiter

Kitchen

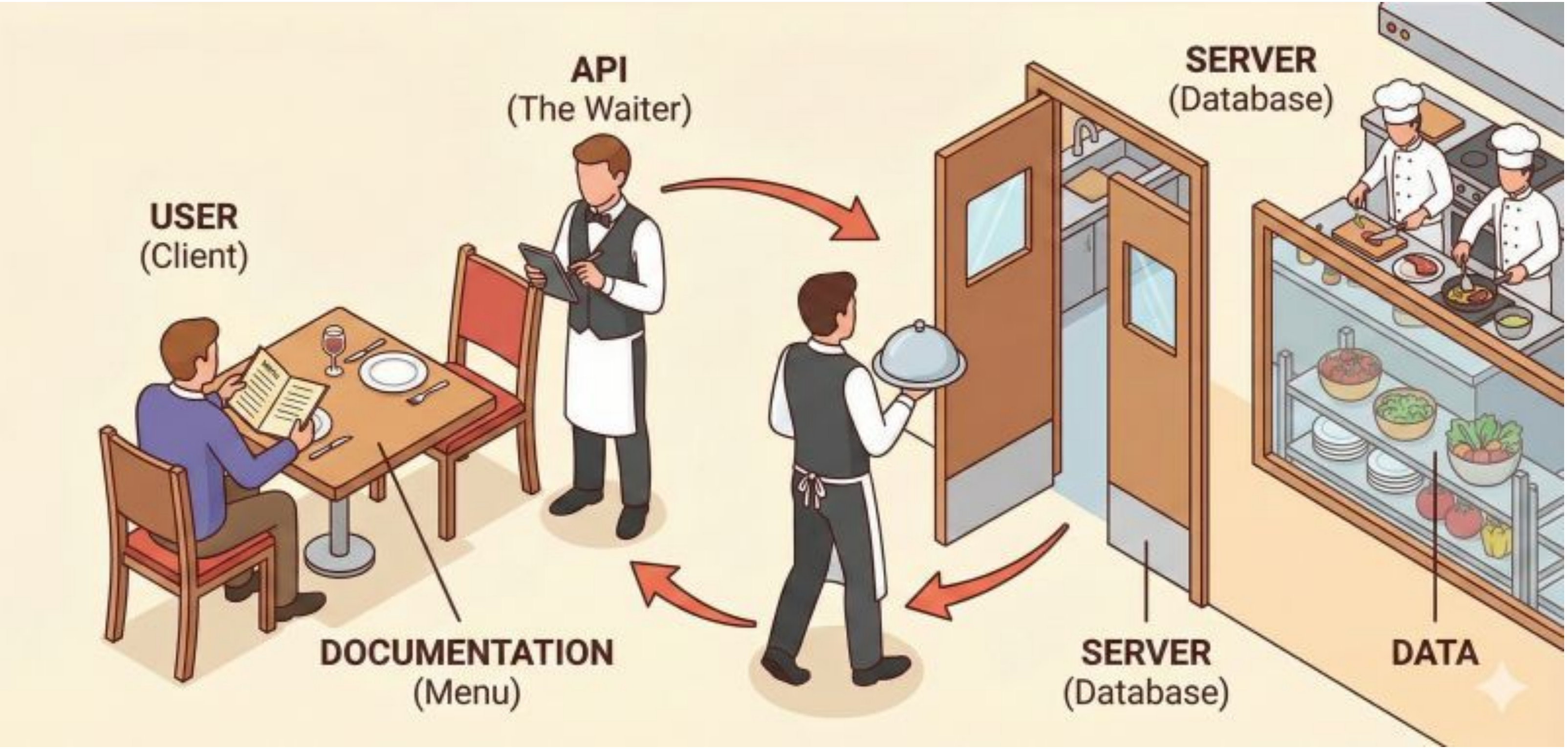
How does a restaurant works?

Client = Python Code

Waiter = API

Kitchen = Server/Database

How does a restaurant works?



https://www.linkedin.com/posts/abdurrahman-tolay-69600323a_api-techeducation-coding-share-7406739405505232896-lLEo/

API WTSS


```
{
    "wtss_version": "4.0.5",
    "links": [
        {
            "rel": "service-doc",
            "type": "text/html",
            "title": "API Documentation",
            "href": "http://data.inpe.br/bdc/wtss/v4/docs"
        },
        {
            "rel": "data",
            "type": "application/json",
            "title": "CBERS4-MUX-2M-1",
            "href": "http://data.inpe.br/bdc/wtss/v4/CBERS4-MUX-2M-1"
        },
        {
            "rel": "data",
            "type": "application/json",
            "title": "CBERS4-WFI-16D-2",
            "href": "http://data.inpe.br/bdc/wtss/v4/CBERS4-WFI-16D-2"
        },
        {
            "rel": "data",
            "type": "application/json",
            "title": "CBERS-WFI-8D-1",
            "href": "http://data.inpe.br/bdc/wtss/v4/CBERS-WFI-8D-1"
        },
        {
            "rel": "data",
            "type": "application/json",
            "title": "LANDSAT-16D-1",
            "href": "http://data.inpe.br/bdc/wtss/v4/LANDSAT-16D-1"
        },
```


https://github.com/brazil-data-cube/wtss-spec

https://data.inpe.br/bdc/wtss/v4/

list_coverages: returns the list of all available coverages in the service.

describe_coverage: returns the metadata of a given coverage.

time_series: query the database for the list of values for a given location and time interval.

API WTSS


```
{
  "coverages": [
    "CBERS4-MUX-2M-1",
    "CBERS4-WFI-16D-2",
    "CBERS-WFI-8D-1",
    "LANDSAT-16D-1",
    "mod11a2-6.1",
    "mod13q1-6.1",
    "myd11a2-6.1",
    "myd13q1-6.1",
    "S2-16D-2"
  ]
}
```


https://data.inpe.br/bdc/wtss/v4/

list_coverages: returns the list of all available coverages in the service.

https://data.inpe.br/bdc/wtss/v4/list_coverages

API WTSS


```
{
    "id": 9,
    "name": "S2-16D",
    "version": "2",
    "fullname": "S2-16D-2",
    "description": "Earth Observation Data Cube generated from Cop
    of spatial resolution, reprojected and cropped to BDC_SM grid
    "title": "Sentinel-2/MSI - Level-2A - Data Cube - LCF 16 days"
    "timeline": [ … ], // 217 items
    "bands": [ … ], // 19 items
    "extent": {
        "type": "Polygon",
        "coordinates": [
            [
                [
                    -74.871069,
                    -34.67556459214432
                ],
                [
                    -74.871069,
                    5.763264005526926
                ],
                [
                    -28.006208041654325,
                    5.763264005526926
                ],
                [
                    -28.006208041654325,
                    -34.67556459214432
                ],
                [
                    -74.871069,
                    -34.67556459214432
                ]
            ]
        ]
    },
```


https://data.inpe.br/bdc/wtss/v4/S2-16D-2

describe_coverage: returns the metadata of a given coverage.

```
{
  "query": {
    "coverage": "S2-16D-2",
    "attributes": [
      "NDVI"
    ],
    "longitude": -49.43414416473638,
    "latitude": -11.819838188733794,
    "start_date": "2025-01-01",
    "end_date": "2025-12-31"
  },
  "result": {
    "attributes": [
      {
        "attribute": "NDVI",
        "values": [
          414,
          586,
          7713,
          1589,
          8163,
          7593,
          7974,
          8229,
          8145,
          7862,
          7700,
          7783,
          8005,
          7859,
          7847,
          7656,
          7561,
          7991,
          7613,
          8152,
          7811,
          6914,
          8325
        ]
      }
    ],
    "timeline": [
      "2025-01-01",
      "2025-01-17",
```


API WTSS

https://data.inpe.br/bdc/wtss/v4/wtss/time_series?coverage=S2-16D-2&attributes=NDVI&latitude=-11.819838188733794&longitude=-49.43414416473638&start_date=2025-01-01&end_date=2025-12-31

time_series: query the database for the list of values for a given location and time interval.

Consuming an API using Python

Consuming an API using Python

```
[7]: import requests

     url = 'https://data.inpe.br/bdc/wtss/v4/'

     r = requests.get(url)

     r
     Last executed at 2026-06-19 14:56:10 in 165ms

[7]: <Response [200]>
```

```
[6]: print(requests.get('https://data.inpe.br/bdc/wtss/v5/'))
     Last executed at 2026-06-19 14:56:07 in 110ms
     <Response [404]>
```

```
[8]: r.content
     Last executed at 2026-06-19 14:56:12 in 7ms
```


```
[8]: b'{"wtss_version":"4.0.5","links":[{"rel":"service-doc","type":"text
     ation/json","title":"CBERS4-MUX-2M-1","href":"http://data.inpe.br/bd
     data.inpe.br/bdc/wtss/v4/CBERS4-WFI-16D-2"},{"rel":"data","type":"ap
     a","type":"application/json","title":"LANDSAT-16D-1","href":"http://
     f":"http://data.inpe.br/bdc/wtss/v4/mod11a2-6.1"},{"rel":"data","typ
     a","type":"application/json","title":"myd11a2-6.1","href":"http://da
     p://data.inpe.br/bdc/wtss/v4/myd13q1-6.1"},{"rel":"data","type":"app
```


```
[9]: r.json()
     Last executed at 2026-06-19 14:56:32 in 11ms
```


```
[9]: {'wtss_version': '4.0.5',
      'links': [{'rel': 'service-doc',
        'type': 'text/html',
        'title': 'API Documentation',
        'href': 'http://data.inpe.br/bdc/wtss/v4/docs'},
       {'rel': 'data',
        'type': 'application/json',
        'title': 'CBERS4-MUX-2M-1',
        'href': 'http://data.inpe.br/bdc/wtss/v4/CBERS4-MUX-2M-1'},
       {'rel': 'data',
        'type': 'application/json',
        'title': 'CBERS4-WFI-16D-2',
        'href': 'http://data.inpe.br/bdc/wtss/v4/CBERS4-WFI-16D-2'},
       {'rel': 'data',
        'type': 'application/json',
```

Python Client

https://github.com/brazil-data-cube/wtss.py

In [ ]:
```
from wtss import *
```

After that, you should create a `wtss` object attached to a given service:

In [ ]:
```
service = WTSS('https://data.inpe.br/bdc/wtss/v4/')
```

The above cell will create an object named `service` that will allow us to comunicate to the given WTSS service.

[4]:
```
service.coverages
```

Last executed at 2026-0

[4]:
```
['CBERS4-MUX-2M-1',
 'CBERS4-WFI-16D-2',
 'CBERS-WFI-8D-1',
 'LANDSAT-16D-1',
 'mod11a2-6.1',
 'mod13q1-6.1',
 'myd11a2-6.1',
 'myd13q1-6.1',
 'S2-16D-2']
```

[5]:
```
service['S2-16D-2']
```

Last executed at 2026-06-19 15:10:55 in 138ms

[5]: **Coverage** S2-16D-2

Description: Earth Observation Data Cube generated from Copernicus Sentinel-2/MSI Level-2A product o
meters of spatial resolution, reprojected and cropped to BDC_SM grid Version 2 (BDC_SM V2), considering

Attributes

name	common name	description	datatype	valid range	scale	nodata
CLEAROB	ClearOb		uint8	{'min': 1.0, 'max': 255.0}	1.0	0.0
TOTALOB	TotalOb		uint8	{'min': 1.0, 'max': 255.0}	1.0	0.0
PROVENANCE	Provenance		int16	{'min': 1.0, 'max': 366.0}	1.0	-1.0
SCL	quality		uint8	{'min': 0.0, 'max': 11.0}	1.0	0.0
B01	coastal		uint16	{'min': 0.0, 'max': 10000.0}	0.0001	0.0
B02	blue		uint16	{'min': 0.0, 'max': 10000.0}	0.0001	0.0
B04	red		uint16	{'min': 0.0, 'max': 10000.0}	0.0001	0.0
B08	nir		uint16	{'min': 0.0, 'max': 10000.0}	0.0001	0.0

Python Client

https://github.com/brazil-data-cube/wtss.py

```
[8]: coverage = service['S2-16D-2']
Last executed at 2026-06-19 15:12:39 in 115ms

[9]: timeline = coverage.timeline

     start = timeline[0]
     end = timeline[-1]

     print(f'Interval range: [{start}, {end}]')
Last executed at 2026-06-19 15:12:39 in 7ms

     Interval range: [2017-01-01, 2026-05-25]
```

```
[10]: ts = coverage.ts(attributes=('B04', 'B08'),
                       latitude=-9.41866, longitude=-61.46103,
                       start_date='2019-01-01', end_date='2019-12-31')
Last executed at 2026-06-19 15:12:52 in 651ms

[11]: ts
Last executed at 2026-06-19 15:12:57 in 20ms
```

[11]: **Time Series** S2-16D-2

B04: [312, 441, 1800, 3154, 294, 901, 333, 285, 143, 167, 129, 183, 148, 747, 745, 674, 673, 708, 415, 1083, 8071, 678, 3695]

B08: [3897, 3964, 3908, 4831, 2759, 2114, 3221, 2362, 2774, 2630, 2638, 2601, 2552, 2356, 2146, 1077, 1035, 1003, 686, 1493, 7741, 1338, 4165]

Scraping

Web Scraping

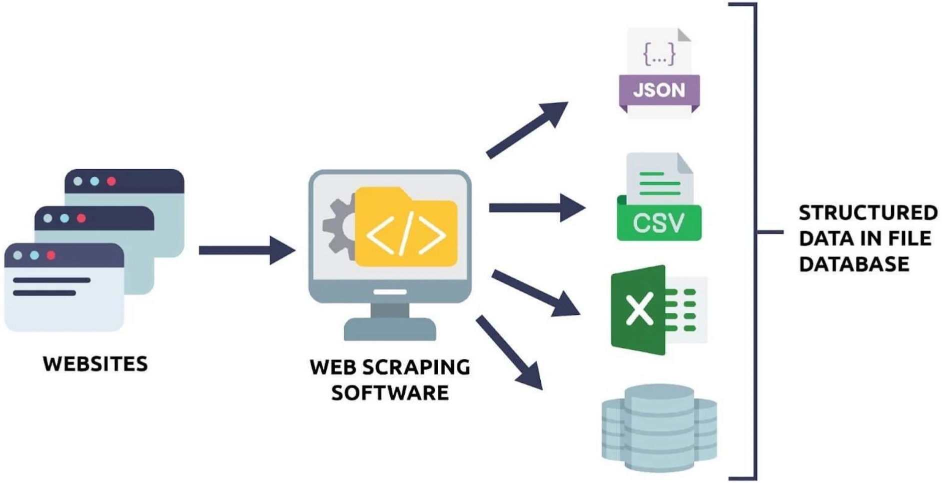
Extract information from Websites

Transform non-structured into structured data

Avoid human error

Faster than copy and paste

Web Scraping



Web Scraping challenges

Keys can repeat

Find a pattern: e.g. first appearance

Sites change

Ip block

Timeouts

Encoding

Web Scraping Tools

requests

BeautifulSoup (HTML)

Scrapy

Selenium

Playwright

Task

GEOINFO

GEOINFO
2026

https://geoinfobr.github.io/index.html

GEOINFO

What would we like to know?

GEOINFO
What would we like to know?

Who are the top publishers considering all editions and each year?

Who publishes in portuguese?

Who are the most frequent visitors?

What is the mean number of authors of a paper?

What are the most common topics?

GEOINFO
What would we like to know?

How does the topics changed through the years?

GEOINFO
What is important to extract?

Title

Author

Date

Abstract

Language

GEOINFO
Scraping Problems

Hyphens

People changing Institution

Repeated names (abbreviations)

Accent (UTF8)

GEOINFO

How to organize this data?

Github

Fork https://github.com/marujore/geoinfo-papers

Comment on an issue (to take that task for you)

Each student will add more information on the repository

Scrap Data from http://mtc-m16d.sid.inpe.br/ibi/8JMKD3MGP7W/3GDK2ML

Format on the same format as the repository

Clear errors

Make a Pull request

Write a short Report on how you have done the task (e-mail)

Web Scraping
Good practices

http://mtc-m16c.sid.inpe.br/robots.txt

```python
from urllib.robotparser import RobotFileParser

def check_robots(url):
    """Check if scraping is allowed"""
    rp = RobotFileParser()
    rp.set_url('http://mtc-m16c.sid.inpe.br/robots.txt')
    rp.read()
    return rp.can_fetch('*', url)

if check_robots('http://mtc-m16c.sid.inpe.br/ibi/8JMKD2USPTW34P/4DKC4FB'):
    print("✅ Scraping allowed")
else:
    print("❌ Scraping blocked by robots.txt")
```

Web Scraping Good practices

Headers

Timeout

Sleep

```
URL = "http://mtc-m16c.sid.inpe.br/col/sid.inpe.br/mtc-m18@

headers = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
        "AppleWebKit/537.36 (KHTML, like Gecko) "
        "Chrome/137.0.0.0 Safari/537.36"
    )
}

response = requests.get(URL, headers=headers, timeout=30)
```

Public Datasets

IBGE

https://servicodados.ibge.gov.br/api/docs/

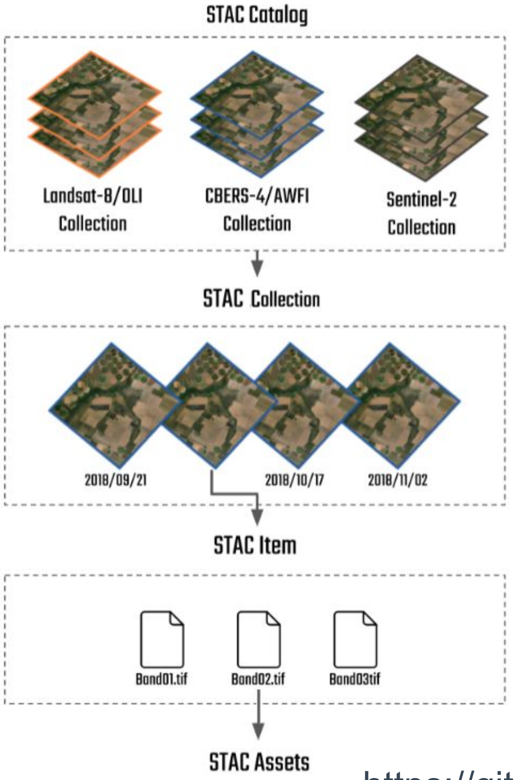
https://sidrapy.readthedocs.io/pt-br/latest/

https://servicodados.ibge.gov.br/api/v1/localidades/municipios/{codigo_ibge}
https://servicodados.ibge.gov.br/api/v1/localidades/municipios/
https://servicodados.ibge.gov.br/api/v1/localidades/municipios/3554805
https://servicodados.ibge.gov.br/api/v4/malhas/municipios/3554805

https://sidrapy.readthedocs.io/pt-br/latest/https:/servicodados.ibge.gov.br/api/v1/localidades/estados/35

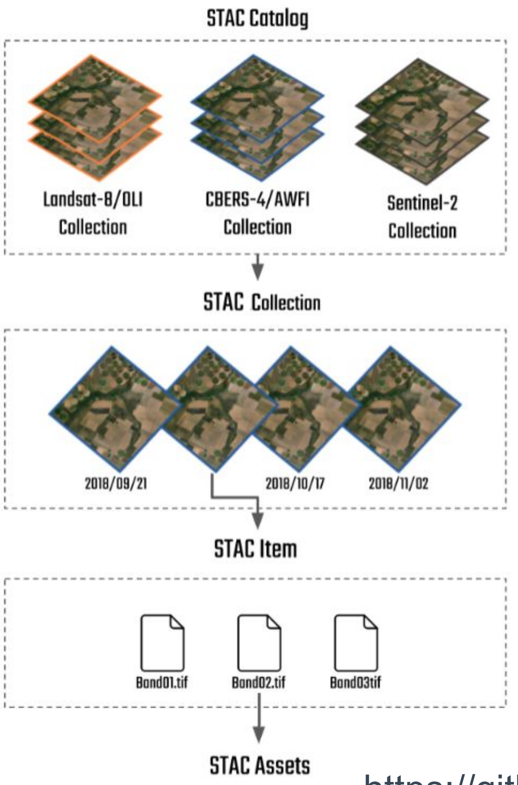
http://www.geoservicos.ibge.gov.br/geoserver/ows?version=1.1.0&

SpatioTemporal Asset Catalog (STAC)



https://github.com/brazil-data-cube/code-gallery

SpatioTemporal Asset Catalog (STAC)



https://github.com/brazil-data-cube/code-gallery

Queimadas

Focos de Queimadas e Incêndios

Arquivos disponibilizados contendo informações sobre focos de queimadas e incêndios florestais em vários intervalos, desde anuais até em "tempo quase real" (a cada 10 minutos). Esses dados estão disponíveis nos formatos CSV (Comma-Separated Values) e KML (Keyhole Markup Language).

ARQUIVOS CSV

ÚLTIMOS 10 MIN
DIÁRIOS
MENSAIS
ANUAIS

ARQUIVOS KML

ÚLTIMOS 3 DIAS
GOES-16 BR
DIÁRIOS · WEB
DIÁRIOS · WEB
Satélite de referência
INSTRUÇÕES
Como acessar via Google Earth

Eventos de Fogo

Integração espaço-temporal de focos de fogo ativo que permite classificar e distinguir incêndios e queimadas no Brasil. Cada evento reúne informações como localização das frentes de fogo, estimativa de área queimada, duração, temperatura máxima do ar e uso do solo. Os dados dos **Eventos Ativos** e **Eventos em Observação** são disponibilizados em formato KML, atualizados a cada hora. *Produto em fase de validação — nível de maturidade Provisório.*

ARQUIVOS KML

EVENTOS ATIVOS
KML
EVENTOS EM OBSERVAÇÃO
KML
INSTRUÇÕES
Como acessar via Google Earth

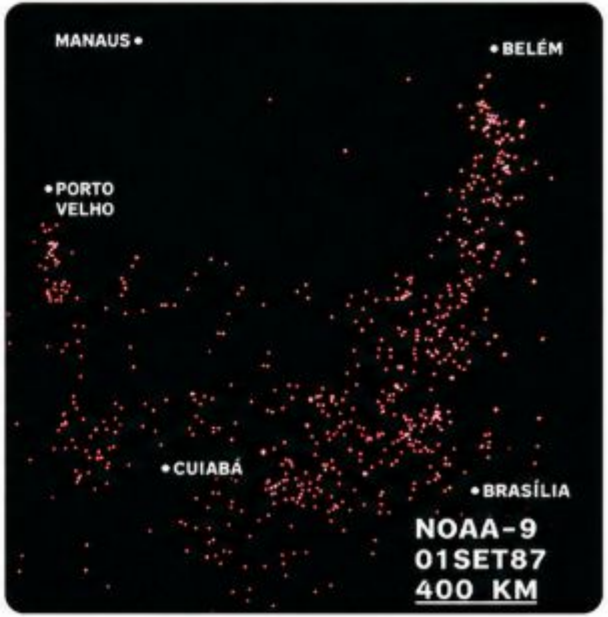
https://data.inpe.br/queimadas/dados-abertos/

https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/10min/

Queimadas

Sistema GE-Image-100 do INPE

Imagem NOAA-9 – Focos de queimadas



À esquerda, o sistema GE-Image-100 do INPE utilizado para o processamento das imagens.
À direita, imagem do satélite NOAA-9 mostrando focos de queimadas na Amazônia.

Dataset Repositories

@flai.inteligencia.artificial

Dataset Repositories

https://www.kaggle.com/datasets

https://data.fivethirtyeight.com/

https://data.nasdaq.com/publishers/QDL

https://dados.gov.br/home

https://data.world/

https://archive.ics.uci.edu/

Task

Github

Issues on

https://github.com/marujore/geoinfo-papers

Material:

Github.io

(to be send)